Exercise -III

1. Let R = {(1, 3), (4, 2), (2, 4), (2, 3), (3, 1)} be a releation on the set A = {1, 2, 3, 4}. The relation R is : **[AIEEE-2004]**

(A) transitive (B) not symmetric (C) reflexive (D) a function

माना कि R = {(1, 3), (4, 2), (2, 4), (2, 3), (3, 1)} समुच्चय A = {1, 2, 3, 4} में एक सम्बन्ध हो, तो सम्बन्ध R है : **[AIEEE-2004]**

(A) संक्रामक (B) सममित नहीं (C) स्वतुल्य (D) एक फलन

Ans. B

Sol. R = {(1, 3), (4, 2), (2, 4), (2, 3), (3, 1)} be a releation on the set A = {1, 2, 3, 4}

for Reflexive : No, because (1, 1), (2, 2) $\notin$ R

for Symmetric : No, because (2, 3) $\in$ R but (3, 2) $\notin$ R

Transitive : No, because every element of Relation R

does not follow (a, b) $\in$ R, (b, c) $\in$ R $\Rightarrow$ (a, c) $\in$R

Option (B)

2. Let R = {(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)}, be relation on the set A = {3, 6, 9, 12}. The relation is : **[AIEEE-2005]**

(A) reflexive and transitive only (B) reflexive only

(C) an equilvalence relation (D) reflexive and symmetric only

माना R = {(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)} समुच्चय A = {3, 6, 9, 12} पर एक सम्बन्ध है। यह सम्बन्ध : **[AIEEE-2005]**

(A) स्वतुल्य तथा संक्रामक है (B) केवल स्वतुल्य है

(C) एक तुल्यता संबंध है (D) स्वतुल्य तथा सममित है

Ans. A

Sol. A = {3, 6, 9, 12}, relation on set A

R = {(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)}

Reflexive : Yes, because (6, 12) $\in$ R but (12, 6) $\notin$ R

(3, 9) $\in$ R but (9, 3) $\notin$ R

(3, 12) $\in$ R but (12, 3) $\notin$ R

(3, 6) $\in$ R but (8, 3) $\notin$ R

Transitive : Yes, becuase elements / ordered pair in Relation R they follow, (a, b) $\in$R, (b, c)$\in$ R $\Rightarrow$ (a, c) $\in$ R

Option (A)

3. Let W denote the words in the English dictionary. Define the relation R by :R = {(x, y) ∈ W × W | the words x and y have at least one letter in common}. Then R is : **[AIEEE-2006]**

(A) reflexive, symmetric and not transitive (B) reflexive, symmetric and transitive

(C) reflexive, not symmetric and transitive (D) not reflexive, symmetric and transitive

माना W अंग्रेजी शब्दकोष में एक शब्द दर्शाता है। सम्बन्ध R निम्न द्वारा परिभाषित है : R = {(x, y) ∈ W × W | शब्दों x तथा y में कम-से-कम एक अक्षर उभयनिष्ठ है, तो R : **[AIEEE-2006]**

(A) स्वतुल्य, सममित पर संक्रामक नही है (B) स्वतुल्य, सममित तथा संक्रामक है

(C) स्वतुल्य, असममित तथा संक्रामक है (D) स्वतुल्य नही, पर सममित तथा संक्रामक है

Ans. A

Sol. W denote the words in the English dictionary

Relation R = {(x, y) ∈ W × W | the word x & y have at tests one letter in common

Reflexive, Yes, x & y have at least one letter in common

Symmetric; Yes, x & y have at least one letter in common

⇒ y & x have at tests are letter in common
Transitive : No,
Let, BAD & CAT have atleast one letter in common
CAT & KITE have at least one letter in common
But BAD & KITE have no letter common
Option (A)

4. Let R be the real line. Consider the following subsets of the plane R × R : S = {(x, y): y = x + 1 and 0 < x < 2} T = {(x, y) : x – y is an integer}. Which one of the following is true? **[AIEEE-2008]**

(A) Both S and T are equivalence relations on R (B) S is an equivalence relation on R but T is not

(C) T is an equivalence relation on R but S is not (D) Neither S nor T is an equivalence relation on R

माना R एक वास्तविक रेखा है। तल R × R के निम्न उपसमुच्चयों को लीजिए
S = {(x, y): y = x + 1 तथा 0 < x < 2}
T = {(x, y) : x – y एक पूर्णांक है}.
निम्न में से कौनसा सत्य है? **[AIEEE-2008]**

(A) S तथा T दोनों R पर तुल्यता सम्बन्ध हैं (B) S , R पर एक तुल्यता सम्बन्ध है लेकिन T नहीं है

(C) T , R पर एक तुल्यता सम्बन्ध है, लेकिन S नहीं है (D) न तो T और न ही S,R पर तुल्यता सम्बन्ध है

Ans. C

Sol. Let R be the real line. Consider the following subsets of the plane R × R

S = {(x, y): y = x + 1
T = {(x, y) : x – y is an integer}
S = (x, y); y = x + 1 & 0 < x < 2
Reflexive : No, because x = x +1

a = 1, can't possible

So no need to check symmetric & Transitive

S $\Rightarrow$ is not a Equivalence relation

T = {(x, y) (x – y)} is an integer

Reflexive : Yes, because x – x = 0 always an integer

Symmetric : Yes, because x – y is an integer then y – x is also an integer

Transitive : yes, because if x – y = Integer .....(i)

y – z = integer ..... (ii)

add equation (i) & (ii) x – z = integer

So T is an equivalence relation

Option (C)

5. If A, B and C are three sets such that $A \cap B = A \cap C$ and $A \cup B = A \cup C$, then : **[AIEEE-2009]**

यदि A, B तथा C तीन ऐसे समुच्चय (sets) हैं जिनके लिए $A \cap B = A \cap C$ तथा $A \cup B = A \cup C$, तब : **[AIEEE-2009]**

(A) A = B (B) A = C (C) B = C (D) $A \cap B = \phi$

Ans. C

Sol. A,B,C are three sets

Such that $A \cap B = A \cap C$ and $A \cup B = A \cup C$

It is only possible when B = C

Option (C) correct

6. Consider the following relations :

R = {(x, y) | x, y are real numbers and x = wy for some rational number w};

$S = \{(\frac{m}{n}, \frac{p}{q}) \mid$ m, n, p and q are integers such that n, q $\neq$ 0 qm = pn.}, Then : **[AIEEE-2010]**

(A) R is an equivalence relation but S is not an equivalence relation

(B) Neither R nor S is an equvalence relation

(C) S is an equvalence relation but R is not an equvalence relation

(D) R and S both are equivalence relations

निम्नलिखित सम्बन्धों को लिजिए

R = {(x, y) | x, y वास्तविक संख्याऐं हैं तथा किसी परिमेय संख्या w के लिए x = wy हैं।

$S = \{(\frac{m}{n}, \frac{p}{q}) \mid$ m, n, p एवं q ऐसे पूर्णांक हैं कि n, q $\neq$ 0 qm = pn.} तो : **[AIEEE-2010]**

(A) R एक तुल्यता सम्बन्ध है लेकिन S एक तुल्यता सम्बन्ध नहीं है

(B) न तो R और नहीं S एक तुल्यता सम्बन्ध है

(C) S एक तुल्यता सम्बन्ध है लेकिन R एक तुल्यता सम्बन्ध नहीं है

(D) R तथा S दोनों ही तुल्यता सम्बन्ध है

Ans. C

Sol. R = {(x, y) | x, y are real numbers and x = wy for some rational number w}

$\because$ x = wy Reflexive x = w. x $\Rightarrow$ w = 1 is rational no.

yes reflexive

Symmetric : If x = wy; then y = wx not always

$\Rightarrow$ R is not equivlence relation

$S = \{(\frac{m}{n}, \frac{p}{q}) \mid$ m, n, p and q are integers such that n, q $\neq 0$ & qm = pn.$\}$

$\because$ qm = pn

$\frac{m}{n} = \frac{p}{q}$

Reflexivel; yest $\frac{m}{n} = \frac{m}{n}$

Symmetric; yes, if $\frac{m}{n} = \frac{p}{q}$ then

Transitive; you, $\frac{m}{n} = \frac{p}{q}, \frac{p}{q} = \frac{r}{s} \Rightarrow \frac{m}{n} = \frac{r}{s}$ ms = rn

$\Rightarrow$ S is an equivalence relation

7. Let R be the set of real numbers : **[AIEEE-2011]**

Statement-1: A = {(x, y) $\in$ R × R : y – x is an integer} is an equivalence relation on R.

Statement :2: B = {(x, y) $\in$ R × R : x = αy for some rational number α} is an equivalence relation on R.

(A) Statement-1 is true, Statement-2 is false

(B) Statement-1 is false, Statement-2 is true

(C) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1

(D) Statement-1 is true, Statement-2 is true; Statement-2 is **not** a correct explanation for Statement-1

माना R वास्तविक संख्याओं का समुच्चय है : **[AIEEE-2011]**

कथन-1: A = {(x, y) $\in$ R × R : y – x एक पूर्णांक ह} R पर एक तुल्यता संबंध है।

कथन-2: B = {(x, y) $\in$ R × R : x = αy किसी परिमेय संख्या α के लिए} R पर एक तुल्यता संबंध है।

(A) कथन-1 सत्य है, कथन -2 असत्य है

(B) कथन-1 असत्य है, कथन -2 सत्य है

(C) कथन-1 सत्य है, कथन-2 सत्य है। कथन-2, कथन-1 की सही व्याख्या है

(D) कथन-1 सत्य है, कथन-2 सत्य हैं, कथन-2 कथन-1 की सही व्याख्या नहीं है

Ans. A

Sol. R be the set of real numbers

$A = \{(x, y) \in R \times R : y - x \text{ is an integer}\}$

Reflexive ; yes because $y - y = 0$ is an integer

Symmetric; yes because $y - x$ is an integer then $x - y$ is also an integer

Transitive; yes because $y - x$ is an integer .....(1)

$x - z$ is an integer .....(2)

add equation (1) & (2) $y - z$ is also an integer

$\Rightarrow$ A is equivalence relation on R

$B = \{(x, y) \in R \times R : x = \alpha y \text{ for some rational number } \alpha\}$

Reflexive; Yes, because $x = \alpha . n \Rightarrow \alpha = 1$ is a rational no.

Option (A)

Symmetric; No, because if $x = \alpha x$ not always is not an equivalence relation on R.

8. Consider the following relation R on the set of real square matrices of order 3.
$R = \{(A, B) \mid A = P^{-1} BP \text{ for some invertible matrix } P\}$.
Statement-1: R is an equivalence relation.
Statement-2: For any two invertible 3×3 matrices M and N, $(MN)^{-1} = N^{-1}M^{-1}$ **[AIEEE-2011]**

(A) Statement-1 is false, statement-2 is true.
(B) Statement-1 is true, statement-2 is true; Statement-2 is correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is true; Statement-2 is not a correct explanation for statement-1.
(D) Statement-1 is true, statement-2 is false.

निम्न सम्बंध R को कोटि 3 के वर्ग आव्यूहों पर विचारित कीजियें :
$R = \{(A, B) \mid A = P^{-1} BP$, किसी व्युत्क्रमणीय आव्यूह P के लिए$P\}$
कथन-1: R एक तुल्यता सम्बंध है।
कथन-2: किन्हीं दो व्युत्क्रमणीय 3×3 आव्यूहों M तथा N के लिए, $(MN)^{-1} = N^{-1}M^{-1}$ **[AIEEE-2011]**

(A) कथन-1 असत्य है और कथन-2 सत्य है
(B) कथन-1 सत्य है और कथन-2 सत्य है। कथन-2 कथन-1 की सही व्याख्या है
(C) कथन-1 सत्य है और कथन-2 सत्य है। कथन-2 कथन-1 की सही व्याख्या नहीं है
(D) कथन-1 सत्य है और कथन-2 असत्य है

Ans. B

Sol. Relation R on the set of real square matrices of order 3.

$R = \{(A, B) \mid A = P^{-1} BP \text{ for some invertible matrix } P\}$.

Reflexive;

9. Let X = {1, 2, 3, 4, 5}. The number of different ordered pairs (Y, Z) that can be formed such that $Y \subseteq X, Z \subseteq X$ and $Y \cap Z$ is empty, is : **[AIEEE-2012]**

माना X = {1, 2, 3, 4, 5} है। इससे निर्मित किए जा सकने वाले क्रमित युग्मों (Y, Z), जो इस प्रकार है कि $Y \subseteq X, Z \subseteq X$ तथा $Y \cap Z$ रिक्त समुच्चय है, कि संख्या है : **[AIEEE-2012]**

(A) 5^3 (B) 5^2 (C) 3^5 (D) 2^5

Ans. C

Sol. X = {1, 2, 3, 4, 5}

$Y \subseteq X, Z \subseteq X$ and $Y \cap Z$ is empty

for $Y \cap Z = \phi$, we have three condition

let a $a \in x$

Case : (1) $a \in y, a \in z$

Case : (2) $a \in y, a \in y$

Case : (3) $a \in z, a \in y$

$Y \cap Z = \phi$ has 3 chances for a

Hence for five elements of X the number of required chance is $3 \times 3 \times 3\ 3 \times 3 = 3^5$

10. Let A and B be two sets containing 2 elements and 4 elements respectively. The number of subsets of $A \times B$ having 3 or more elements is : **[JEE(Main)2013]**

माना A तथा B दो ऐसे सम्मुच्य है जिनमें क्रमशः 2 अवयव तथा 4 अवयव है। $A \times B$ के उन उपसमुच्चयों की संख्या जिनमें 3 अथवा अधिक अवयव हैं, है : **[JEE(Main)2013]**

(A) 256 (B) 220 (C) 219 (D) 211

Ans. C

Sol. n (A) = 2, n (B) = 8

The number of subsets of $A \times B$ having 3 or more elements is

$A \times B$, has me subsets with zero element = 1

$A \times B$, has number of subsets with one elements = $8_{C_1} = 8$

$A \times B$, has number of subsets with two elements = $8_{C_2} = 28$

Total number of subsets = $2^8 = 250$

Required number of subsets = 256 – (28 + 8 +1)

= 219

11. Let $S = \{x \in R: x \geq 0 \text{ and } 2|\sqrt{x}-3| + \sqrt{x}(\sqrt{x}-6)+6=0\}$ Then S : **[JEE(Main)2018]**

(A) contains exactly two elements (B) contains exactly four elements

(C) is an empty set (D) contains exactly one element

माना $S = \{x \in R: x \geq 0$ तथा $2|\sqrt{x}-3| + \sqrt{x}(\sqrt{x}-6)+6=0\}$ तो S : **[JEE(Main)2018]**

(A) में मात्र दो अवयव हैं (B) में मात्र चार अवयव हैं

(C) एक रिक्त समुच्चय हैं (D) में मात्र एक ही अवयव हैं

Ans. A

Sol. $S = \{x \in R: \geq 0 \text{ and } 2|\sqrt{x}-3| + \sqrt{x}(\sqrt{x}-6)+6=0\}$

Case : (1) $\sqrt{x}-3 \geq 0$

$\sqrt{x} \geq 3$

$x < a$

$\Rightarrow 2\left(\sqrt{x}-3\right)+x-6\sqrt{x}+6=0$

$2\sqrt{x}-6+x-6\sqrt{x}+6=0$

$x-4\sqrt{x}=0$

$\sqrt{x}\left(\sqrt{x}-4\right)=0$

$\sqrt{x}=0, x=0$ or $\sqrt{x}-0=0$

$\sqrt{x}=4$

$x=16$

$\Rightarrow x=0$ Rejected

$x=16$, accept $x \geq a$

Case : (2) $\sqrt{x}-3<0$ but $x \geq 0$

$\sqrt{x}<3$

$x<9,\ x \geq a$

$0 \leq x < a$

$\Rightarrow -2\left(\sqrt{x}-3\right)+x-6\sqrt{x}+6=0$

$-2\sqrt{x}+6+z-6\sqrt{x}+6=0$

$x-2\sqrt{x}+12=0$

Let $\sqrt{x}=t$

$t^2-8t+12=0$

$(t-6)(t-2)=0$

$t=6, t=2$

$\sqrt{x}=6,\ \sqrt{x}=2$

$x=36, x=4$

$x=36$ reject

$x=4$ accept $(0 \leq x < a)$

$x=16, x=4$ conains exactly two elements

12. Let $A = \{n \in N : n \text{ is a 3–digit number}\}$
$B = \{9k + 2 : k \in N\}$
and $C = \{9k + l : k \in N\}$ for some l $(0 < l < 9)$
If the sum of all the elements of the set $A \cap (B \cup C)$ is 274×400, then l is equal to _____ .
[JEE(Main)Feb.2021]

माना $A = \{n \in N : n$ एक 3–अंकों की संख्या है$\}$
$B = \{9k + 2 : k \in N\}$
तथा $C = \{9k + l : k \in N\}$ किसी l $(0 < l < 9)$ के लिए।
यदि समुच्चय $A \cap (B \cup C)$ के सभी अवयवों का योग 274×400 है, तो l बराबर है _____
[JEE(Main)Feb.2021]

Ans. 5

Sol. If $B \cup C = B$

$$\sum_{K=11}^{110}(9k+2) = 9\left(\sum_{K=1}^{110}K - \sum_{K=1}^{10}K\right) + 200$$

$= 54650 \neq 274 \times 400$

If $B \cup C \neq B$

$$\sum_{K=11}^{110}(9k+2) + \sum_{K=11}^{110}(9k+\ell) = 274 \times 400$$

$\ell = 5$

13. Let $R = \{(P, Q) \mid P$ and Q are at the same distance from the origin$\}$ be a relation, then the equivalence class of $(1, -1)$ is the set : **[JEE(Main)Feb.2021]**

माना $R = \{(P, Q) \mid P$ तथा Q, मूलबिंदु से समान दूरी पर हैं$\}$ तो $(1, -1)$ का तुल्यता – वर्ग निम्न में से कौन सा समुच्चय है ?
[JEE(Main)Feb.2021]

(A) $S = \left\{(x, y) \mid x^2 + y^2 = \sqrt{2}\right\}$ (B) $S = \left\{(x, y) \mid x^2 + y^2 = 2\right\}$

(C) $S = \left\{(x, y) \mid x^2 + y^2 = 4\right\}$ (D) $S = \left\{(x, y) \mid x^2 + y^2 = 1\right\}$

Ans. B

Sol. R $(1, -1)$ P (h, k), O $(0, 0)$

$OR = OP$

locus of P is $h^2 + k^2 = 2$

$x^2 + y^2 = 2$

14. Define a relation R over a class of $n \times n$ real matrices A and B as "ARB iff there exists a non–singular matrix P such that $PAP^{-1} = B$" Then which of the following is true? **[JEE(Main)Mar.2021]**

(A) R is reflexive, transitive but not symmetric (B) R is an equivalence relation

(C) R is symmetric, transitive but not reflexive (D) R is reflexive, symmetric but not transitive

$n \times n$ के वास्तविक आव्यूहों A तथा B के एक समूह पर एक संबंध R निम्न प्रकार से परिभाषित है : 'ARB' यदि और केव ल यदि एक व्युत्क्रमणीय आव्यूह P का अस्तित्व है जिनके लिए $PAP^{-1} = B$ है" । तो निम्न में से कौनसा– सत्य है?

[JEE(Main)Mar.2021]

(A) R स्वतुल्य और संक्रामक है परन्तु सममित नहीं है (B) R एक तुल्यता संबंध है

(C) R स्वतुल्य और सममित है परन्तु संक्रामक नहीं है (D) R सममित और संक्रामक है परन्तु स्वतुल्य नहीं है

Ans. B

Sol. For reflexive

$ARA \Rightarrow PAP^{-1} = A$ ..... (1) must be true

For $P = I$, eq. (1) is true so 'R' is reflexive.

For symmetric

$ARB \Leftrightarrow PAP^{-1} = B$ ..... (1) is true

For BRA iff $PBP^{-1} = A$.... (2) must be true

$\because PAP^{-1} = B$

$P^{-1} PA P^{-1} = P^{-1}B$

$I A P^{-1}P = P^{-1}BP$

$A = P^{-1}BP$ .... (3)

From (2) and (3) $PBP^{-1} = P^{-1}BP$ can be true for some $P = P^{-1}$

$\Rightarrow P^2 = I$ (det (P) $\neq$ 0)

So, R is symmetric

For transitive

$ARB \Leftrightarrow PAP^{-1} = B$ ..... is true

$BRC \Leftrightarrow PBP^{-1} = C$ ..... is true

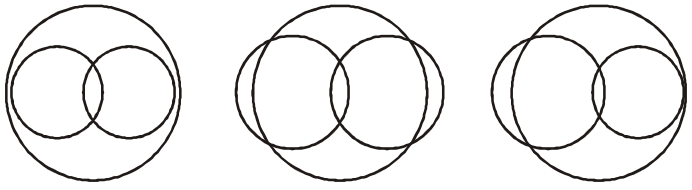
Now $PPA P^{-1} P^{-1} = C$

$P^2A (P^2)^{-1} = C \Leftrightarrow ARC$

So R is transitive relation

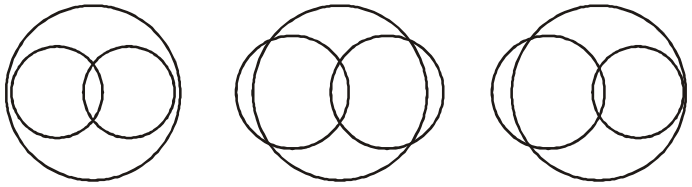
$\Rightarrow$ Hence, R is equivalence

15. In a school, there are three types of games to be played. Some of the students play two types of games, but none play all the three games. Which Venn diagrams can justify the above statement ? **[JEE(Main)Mar.2021]**



(A) P and Q (B) Q and R (C) None of these (D) P and R

एक विद्यालय में तीन प्रकार के खेल खेले जाते है। कुछ विद्यार्थी दो प्रकार के खेल खेलते हैं परन्तु कोई भी विद्यार्थी तीनों खेल नहीं खेलता है। निम्न में से कौनसा वेन आरेख उपरोक्त कथन को प्रदर्शित करता है? **[JEE(Main)Mar.2021]**



P Q R

(A) P तथा Q (B) Q तथा R (C) इनमें से कोई नहीं (D) P तथा R

Ans. C

Sol. None of P, Q, R represent the exact condition given in the problem. So option (3) is right answer

16. Let A = {2, 3, 4, 5,...., 30} and '$\simeq$' be an equivalence relation on A × A, defined by (a, b) $\simeq$ (c, d), if and only if and only if ad = bc. Then the number of ordered pairs which satisfy this equivalence relation with ordered pair (4, 3) is equal to : **[JEE(Main)Mar.2021]**

माना A = {2, 3, 4, 5,...., 30} है तथा A × A पर, (a, b) $\simeq$ (c, d), यदि और केवल यदि ad = bc है, द्वारा परिभाषित एक तुल्यता संबंध '$\simeq$' है। तो क्रमित युग्मों की संख्या, जो क्रमित युग्म (4, 3) के साथ इस तुल्यता संबंध को संतुष्ट करते हैं, है : **[JEE(Main)Mar.2021]**

(A) 5 (B) 8 (C) 6 (D) 7

Ans. D

Sol. $(a,b) \simeq (c,d) \Leftrightarrow \frac{a}{b} = \frac{c}{d}$

$\frac{4}{3} = \frac{8}{6} = \frac{12}{9} = \frac{16}{12} = \frac{20}{15} = \frac{24}{18} = \frac{28}{21}$

Ordered pairs (4, 3), (8, 6), (12, 9), (16, 12), (20, 15), (24, 18), (28, 21)

Total = 7

17. The number of elements in the set $\{x \in R: (|x| - 3)|x + 4| = 6\}$ is equal to : **[JEE(Main)Mar.2021]**

समुच्चय $\{x \in R: (|x| - 3)|x + 4| = 6\}$ में अवयवों की संख्या है : **[JEE(Main)Mar.2021]**

(A) 2 (B) 1 (C) 3 (D) 4

Ans. A

Sol. $(|x| - 3)|x + 4| = 6$

Case - I $x \geq 0$

$(x - 3)(x + 4) = 6$

$x^2 + x - 12 = 6$

$x^2 + x - 18 = 0$

$x = \frac{-1 \pm \sqrt{173}}{2} \quad \Rightarrow x = \frac{1 + \sqrt{73}}{2} > 0$

Case - II $-4 \leq x < 0$

$(-x - 3)(x + 4) = 6$

$x^2 + 7x + 12 - 6$

$x^2 + 7x + 18 = 0 \quad D < 0$

Case - III $x < -4$

$(-x-3)(-x-4) = 6$

$x^2 + 7x + 12 = 6$

$x^2 + 7x + 6 = 0$

$(x+1)(x+6) = 0$

$x = -1, x = -6 \quad \Rightarrow \quad x = -6$

18. Let N be the set of natural numbers and a relation R on N be defined by $R = \{(x, y) \in N \times N : x^3 - 3x^2y - xy^2 + 3y^3 = 0\}$. Then the relation R is : **[JEE(Main)July2021]**

(A) reflexive but neither symmetric nor transitive

(B) symmetric but neither reflexive nor transitive

(C) an equivalence relation

(D) reflexive and symmetric, but not transitive

माना N प्राकृतिक संख्याओं का समुच्चय है और N पर एक संबंध R निम्न द्वारा परिभाषित है

$R = \{(x, y) \in N \times N : x^3 - 3x^2y - xy^2 + 3y^3 = 0\}$ । तो संबंध R : **[JEE(Main)July2021]**

(A) स्वतुल्य है, परन्तु न तो स्वतुल्य है और न ही संक्रामक है

(B) सममित है, परन्तु न तो स्वतुल्य है और न ही संक्रामक है

(C) एक तुल्यता संबंध है

(D) स्वतुल्य और सममित है, परन्तु संक्रामक नहीं है

Ans. A

Sol. $(x, y) \in N \times N$

$x^3 - 3x^2y - xy^2 + 3y^3 = 0$

$\Rightarrow \quad x(x^2 - y^2) - 3y(x^2 - y^2) = 0$

$\Rightarrow \quad (x - 3y)(x^2 - y^2) = 0$

$\Rightarrow \quad (x - y)(x + y)(x - 3y) = 0$

$\therefore \; x = y \text{ or } x = -y \text{ or } x = 3y$

It is a reflexive relation Since xRx

$\because \; x = 3y$

Consider $(3, 1) \in R$

But $(1, 3) \notin R$

$\because$ Hence is is not Symmetric

For transitive Consider $(9, 3) \in R$ and $(3, 1) \in R$

but $(9, 1)) \notin R$

$\therefore$ It is not transitive.

19. Let $A = \{n \in N\} \, |n^2 \leq +10{,}000\}$, $B = \{3k + 1\} \mid k \in N\}$ and $C = \{2k \mid k \in N\}$, then the sum of all the elements of the set $A \cap (B - C)$ is equal to _____. **[JEE(Main)July2021]**

माना $A = \{n \in N\} | n^2 \leq +10,000\}$, $B = \{3k+1\} | k \in N\}$ तथा $C = \{2k | k \in N\}$ हैं, तो समुच्चय $A \cap (B - C)$ के सभी अवयवों का योगफल बराबर है _____ **[JEE(Main)July2021]**

Ans. 832

Sol. $A = \left\{n \in N \mid n^2 \leq n + 1000\right\}$

$B = \left\{3k+1 \mid k \in N\right\}$

$C = \left\{2k \mid k \in N\right\}$

$\therefore$ For $n^2 \leq n + 10000$

$\Rightarrow n(n-1) \leq 10000$

$\therefore$ Product of two consecutive nature No. ≤ 10000

$\therefore A = \{1, 2, 3, 4, ...,100\}$

$\therefore B = \{4, 7, 10 .....97, 100, 103.....\}$

$C = \{2, 4, 6, 8, ...98, 100, 102.....\}$

$\therefore B - C = \{7, 13, 19, ...91, 97\}$

$A \cap (B - C) = \{7, 13, 19 .....91, 97, \}$

No. or elements = 16

Sum of all the elements in $A \cap (B - C)$

$= 7 + 13 + 19 + ...+ 91 + 97$

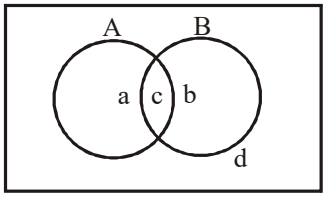
$= \frac{16}{2}[7+97] = 16 \times 52 = 832$

20. Out of all the patients in a hospital 89% are found to be suffering from heart ailment and 98% are suffering from lungs infection. If K% of them are suffering from both ailments, then K can not belong to the set : **[JEE(Main)Aug.2021]**

एक अस्पताल के सभी मरीजों में से 89% दिल की बीमारी से ग्रसित पाये गये तथा 98% के फेफड़े संक्रमित पाये गये। यदि K% दोनों बीमारियों से ग्रसित हैं, तो निम्न में से किस समुच्चय में K नहीं हो सकता? : **[JEE(Main)Aug.2021]**

(A) {84, 87, 90, 93} (B) {79, 81, 83, 85} (C) {84, 86, 88, 90} (D) {80, 83, 86, 89}

Ans. B

Sol. 

$A \equiv$ Heart ailment ; $B \equiv$ lung ailment

$a + b + c + d = 100$ .....(1)

$a + c = 89$ .....(2)

$b + c = 98$ .....(3)

$a + b + 2c = 187$

$a + b + c \le 100$

$\Rightarrow c \ge 87$

21. Which of the following is not correct for relation R on the set of real numbers? : **[JEE(Main)Aug.2021]**

(A) $(x, y) \in R \Leftrightarrow 0 < |x| - |y| \le 1$ is neither transitive nor symmetric

(B) $(x, y) \in R \Leftrightarrow |x - y| \le 1$ is reflexive and symmetric

(C) $(x, y) \in R \Leftrightarrow 0 < |x - y| \le 1$ is symmetric and tansitive

(D) $(x, y) \in R \Leftrightarrow |x| - |y| \le 1$is reflexive but not symmetric

निम्न में से कौनसा वास्तविक संख्याओं के समुच्चय पर संबंध R के लिए सही नहीं है ? : **[JEE(Main)Aug.2021]**

(A) $(x, y) \in R \Leftrightarrow 0 < |x| - |y| \le 1$ स्वतुल्य है किन्तु सममित नहीं है

(B) $(x, y) \in R \Leftrightarrow |x - y| \le 1$ स्वतुल्य तथा सममित है

(C) $(x, y) \in R \Leftrightarrow 0 < |x - y| \le 1$ सममित तथा संक्रामक है

(D) $(x, y) \in R \Leftrightarrow |x| - |y| \le 1$ न तो संक्रामक है न ही सममित है

Ans. C

Sol. $(x, y) \in R \Leftrightarrow 0 < |x - y| \le 1$ is symmetric but not transitive.

22. If $A = \{x \in R : |x-2| > 1\}$, $B = \left\{x \in R : \sqrt{x^2 - 3} > 1\right\}$, $C = \{x \in R : |x-4| \ge 2\}$ and Z is the set of all integers, then the number of subsets of the set $(A \cap B \cap C)^c \cap Z$ is_____. **[JEE(Main)Aug.2021]**

यदि $A = \{x \in R : |x - 2| > 1\}$, $B = \left\{x \in R : \sqrt{x^2 - 3} > 1\right\}$, $C = \{x \in R : |x-4| \ge 2\}$ है तथा सभी पूर्णांको का समुच्चय Z है, तो समुच्चय $(A \cap B \cap C)^c \cap Z$ के उपसमुच्चयों की संख्या है_____ **[JEE(Main)Aug.2021]**

Ans. 256

Sol. $A : x - 2 < -1$ or $x - 2 > 1$

$x < 1$ or $x > 3$

$x \in (-\infty, 1) \cup (3, \infty)$

$B : \sqrt{x^2 - 3} > 1 \Rightarrow x^2 - 3 > 1$

$\Rightarrow x^2 > 4$

$\Rightarrow x \in (-\infty, -2) \cup (2, \infty)$

$C : x - 4 \le -2$ or $x - 4 \ge 2$

$\Rightarrow x \le 2$ or $x \ge 6$

$\Rightarrow x \in (-\infty, 2] \cup [6, \infty)$

$A \cap B \cap C = (-\infty, -2) \cup [6, \infty)$

$(A \cap B \cap C)^C = [-2, 6)$

$(A \cap B \cap C)^C \cap Z = \{-2, -1, 0, 1, 2, 3, 4, 5\}$

Number of subsets $= 2^8 = 256$

23. Let $A = \{x \in R : |x + 1| < 2\}$ and $B = \{x \in R : |x - 1| \geq 2\}$. Then which one of the following statements is NOT true? **[JEE(Main)June2022]**

माना $A = \{x \in R : |x + 1| < 2\}$ तथा $B = \{x \in R : |x - 1| \geq 2\}$ हैं। तो निम्न कथनों में से कौनसा एक सत्य नहीं है : **[JEE(Main)June2022]**

(A) $A - B = (-1, 1)$ (B) $B - A = R - (-3, 1)$

(C) $A \cap B = (-3, -1]$ (D) $A \cup B = R - [1, 3)$

Ans. B

Sol. A: $|x + 1| < 2$

$-2 < x + 1 < 2$

$x \in (-3, 1)$

B $|x - 1| \geq 2$

$x - 1 \in (-\infty, -2] \cup [2, \infty)$

$x \in (-\infty, -1] \cup [3, \infty)$

$A - B = (1, 1)$

$B - A = (-\infty, -3] \cup [3, \infty) = R - (-3, 3)$

$A \cap B = (-3, -1]$

$A \cup B = P(-\infty, 1) \cup [3, \infty) = R - [1, 3)$

24. Let $R_1 = \{(a, b) \in N \times N : |a - b| \leq 13\}$ and $R_2 = \{(a, b) \in N \times N : |a - b| \neq 13\}$. Then on N : **[JEE(Main)June2022]**

(A) Both R_1 and R_2 are equivalence relations

(B) Neither R_1 nor R_2 is an equivalence relation

(C) R_1 is an equivalence relation but R_2 is not

(D) R_2 is an equivalence relation but R_1 is not

माना $R_1 = \{(a, b) \in N \times N : |a - b| \leq 13\}$ तथा $R_2 = \{(a, b) \in N \times N : |a - b| \neq 13\}$ हैं। तो N पर : **[JEE(Main)June2022]**

(A) R_1 तथा R_2 दोनों तुल्यता संबंध हैं

(B) न तो R_1 ही R_2 एक तुल्यता संबंध है

(C) R_1 एक तुल्यता संबंध है परन्तु R_2 नहीं है

(D) R_2 एक तुल्यता संबंध है परन्तु R_1 नहीं है

Ans. B

Sol. $R_1 = \{(a, b) \in N \times N : |a - b| \leq 13\}$ and $R_2 = \{(a, b) \in N \times N : |a - b| \neq 13\}$ in R_1 for transitive let 3 numbers a = 1, b = 12, c = 19

$\because$ $|a - b| \le 13$

$\therefore$ $|1 - 12| \le 13 \Rightarrow (1, 12) \in R_1$,

similarly $(12, 19) \in R_1$ but $(1, 19) \notin R_1$. So R_1 is not transitive. Therefore R_1 is not equivalance.

In R_2 for transistive let 3 numbers.

$a = 10$, $b = 15$, $c = 23$

$\because$ $|a - b| \ne 13$

$\therefore$ $|10 - 15| \ne 13 \Rightarrow (10, 15) \in R_2$, Similarly $(15, 23) \in R_2$ but $(10, 23) \in R_2$

So R_2 is not transitive

Therefore R_2 is not equivalance

25. Let a set $A = A_1 \cup A_2 \cup ..... \cup A_k$, where $A_i \cap A_j = \phi$ for $i \ne j$, $1 \le i, j \le k$. Define the relation R from A to A by $R = \{(x, y) : y \in A_i$ if and only if $x \in A_i, 1 \le i \le k\}$. Then, R is : **[JEE(Main)June2022]**

(A) reflexive, symmetric but not transitive

(B) reflexive, transitive but not symmetric

(C) reflexive but not symmetric and transitive

(D) an equivalence relation

माना एक समुच्चय $A = A_1 \cup A_2 \cup ..... \cup A_k$ हैं। जहाँ $A_i \cap A_j = \phi$, $i \ne j$, $1 \le i, j \le k$ है। A से में संबंध R, $R = \{(x, y) : y \in A_i$ यदि और केवल यदि $x \in A_i, 1 \le i \le k\}$: **[JEE(Main)June2022]**

(A) स्वतुल्य तथा सममित है परन्तु संक्रामक नहीं है

(B) स्वतुल्य तथा संक्रामक है परन्तु सममित नहीं है

(C) स्वतुल्य है परन्तु सममित तथा संक्रामक नहीं है

(D) एक तुल्यता संबंध है

Ans. D

Sol. $A = A_1 \cup A_2 \cup A_3 A ___ \cup A_k$

$A_i \cap A_i = \phi \quad i \ne j$

$R : R \to A \qquad 1 \le i_j j \le k$

1. $R \equiv \{(x, x) \; \forall \; x \in A\}$ R is an equivdence relation

2. Symmetric if possible let $(x, y) \in R \;\; x, y \in A$;

$\Rightarrow y \in A_i$ iff $x \in A_i \;\; 1 \le i \le k$

$\Rightarrow x_f \in A_i$ iff $y \in A_i \;\; 1 \le i \le k$

$\Rightarrow (y, x) \in R$

$(x, y) \in R \Rightarrow (y, x)$ ∄$x, y \in A$

$\Rightarrow$ R is symmetric

3. Transitive if possible let $(x,y),(y,z) \in R\ x,y,z \in A_i$

$\Rightarrow (x,y) \in R$ and $(y,z) \in R$

$\Rightarrow y \in A_i$ iff $x_i \in A_i$ and $z \in A_i$ iff $y \in A_i$

$\Rightarrow x_i \in A_i$ iff $y_i \in A_i$ and $z \in A_i$ iff $y \in A_i$

$\Rightarrow Z \in A_i$ iff $x \in A_i$

$\Rightarrow (x,z) \in R$

$(x,y),(y,z) \in R \Rightarrow (x,z) \in R \quad \forall\ x,y,z \in A_i$

R is transitive

Hence R is an equivalence redation

26. Let A = {n ∈ N : H.C.F. (n, 45) = 1} and Let B = {2k : k ∈ {1, 2,...,100}}. Then the sum of all the elements of $A \cap B$ is _____. **[JEE(Main)June2022]**

माना A = {n ∈ N : H.C.F. (n, 45) = 1 है} तथा माना B = {2k : k ∈ {1, 2,...,100}} है। तो $A \cap B$ के सभी अवयवों का योग है _____ **[JEE(Main)June2022]**

Ans. 5264

Sol. Sum of elements in $A \cap B$

$$= \underbrace{(2+4+6+...+200)}_{\text{multiple of 2}} - \underbrace{(6+12+...+198)}_{\text{multiple of 2 \& 3 i.e. 6}} - \underbrace{(10+20+...+200)}_{\text{multiple of 5 \& 2 i.e. 10}} + \underbrace{(30+60+...+180)}_{\text{multiple of 2,5 \& 3 i.e. 30}}$$

= 5264

27. Let R_1 and R_2 be relations on the set {1, 2, ..., 50} such that

$R_1 = \{(p, p^n) : p \text{ is a prime and } n \geq 0 \text{ is an integer}\}$ and

$R_2 = \{(p, p^n) : p \text{ is a prime and } n = 0 \text{ or } 1\}$

Then, the number of elements in $R_1 - R_2$ is _____. **[JEE(Main)June2022]**

माना समुच्चय 1, 2, ..., 50} पर दो संबंध R_1 तथा R_2 निम्न प्रकार हैं :

$R_1 = \{(p, p^n) : p$ एक अभाज्य संख्या है तथा $n \geq 0$ एक पूर्णांक} तथा

$R_2 = \{(p, p^n) : p$ एक अभाज्य संख्या है तथा $n = 0$ या 1 है}

तो $R_1 - R_2$ में अवयवों की संख्या है _____ **[JEE(Main)June2022]**

Ans. 8

Sol. Here, $p, p^n \in \{1, 2,...50\}$

Now p can take values

2,3,5,7,11,13,17,23,29,31,37,41,43 and 47.

$\therefore$ we can calculate no. of elements in R, as

$(2, 2°), (2,21) .. (2,2^5)$

$(3, 3^\circ)$, ... $(3, 3^3)$

$(5, 5^\circ)$, ... $(5, 5^2)$

$(7, 7^\circ)$, ... $(7, 7^2)$

$(11, 11^\circ)$, ... $(11, 11^1)$

And rest for all other two elements each

$\therefore n(R_1) = 6 + 4 + 3 + 3 + (2 \times 10) = 36$

Similarly for R_2

$(2, 2^\circ), (2, 2^1)$

$(47, 47^\circ), (47, 47^1)$

$\therefore n(R_2) = 2 \times 14 = 28$

$\therefore n(R_1) - n(R_2) = 36 - 28 = 8$

28. Let R_1 and R_2 be two relations defined on R by $a\ R_1\ b \Leftrightarrow ab \geq 0$ and $a\ R_2\ b \Leftrightarrow a \geq b$. Then :

[JEE(Main)July2022]

(A) R_1 is an equivalence relation but not R_2 (B) R_2 is an equivalence relation but not R_1

(C) Both R_1 and R_2 are equivalence relations (D) Neither R_1 nor R_2 is an equivalence relation

माना R पर दो संबध R_1 तथा R_2 , $a\ R_1\ b \Leftrightarrow ab \geq 0$ तथा $a\ R_2\ b \Leftrightarrow a \geq b$ द्वारा परिभाषित हैं। तो :

[JEE(Main)July2022]

(A) R_1 एक तुल्यता संबध है परन्तु R_2 नहीं है (B) R_2 एक तुल्यता संबध है परन्तु R_1 नहीं है

(C) R_1 तथा R_2 दोनों तुल्यता संबध हैं (D) न तो R_1 न ही R_2 एक तुल्यता संबध है

Ans. D

Sol. $R_1 = \{xy \geq 0, x, y \in R\}$

For reflexive $x \times x \geq 0$ which is true.

For symmetric

If $xy \geq 0 \Rightarrow yx \geq 0$

If $x = 2, y = 0$ and $z = -2$

Then $x.y \geq 0$ & $y.z \geq 0$ but $x.z \geq 0$ is not true

$\Rightarrow$ not transitive relation.

$\Rightarrow R_1$ is not equivalence

R_2 if $a \geq b$ it does not implies $b \geq a$

$\Rightarrow R_2$ is not equivalence relation

$\Rightarrow$ D

29. For $\alpha \in N$, consider a relation R on N given by $R = \{(x, y) : 3x + \alpha y$ is a multiple of $7\}$. The relation R is an equivalence relation if and only if : **[JEE(Main)July2022]**

(A) $\alpha = 14$

(B) α is a multiple of 4

(C) 4 is the remainder when α is divided by 10

(D) 4 is the remainder wehn α is devided by 7

$\alpha \in N$ के लिए, N पर एक संबंध R,R = {(x, y) : 3x + αy, 7 का एक गुणज है} द्वारा दिया गया है। संबंध R एक तुल्यता संबंध है यदि और केवल यदि : **[JEE(Main)July2022]**

(A) $\alpha = 14$ है

(B) α, 4 का एक गुणज है

(C) α को 10 से विभाजित करने पर शेषफल 4 है

(D) α को 7 से विभाजित करने पर शेषफल है 4 है

Ans. D

Sol. For R to be reflexive $\Rightarrow$ x R x

$\Rightarrow 3x + \alpha x = 7x \Rightarrow (3 + \alpha) x = 7K$

$\Rightarrow 3 + \alpha = 7\lambda \Rightarrow \alpha = 7\lambda - 3 = 7N + 4$, K, λ,N $\in$ I

$\therefore$ when α divided by 7, remainder is 4.

R to be symmetric xRy $\Rightarrow$ yRx

$3x + \alpha y = 7N_1$, $3y + \alpha x = 7N_2$

$\Rightarrow (3 + \alpha)(x + y) = 7(N_1 + N_2) = 7N_3$

Which holds when $3 + \alpha$ is multiple of 7

$\therefore \alpha = 7N + 4$ (as did earlier)

R to be transitive

xRy&yRz $\Rightarrow$ xRz .

$3x + \alpha y = 7N_1$ & $3y + \alpha z = 7N_2$ and

$3x + \alpha z = 7N_3$

$\therefore 3x + 7N_2 - 3y = 7N_3$

$\therefore 7N_1 - \alpha y + 7N_2 - 3y = 7N_3$

$\therefore 7(N_1 + N_2) - (3 + \alpha) y = 7N_3$

$\therefore (3 + \alpha) y = 7N$

Which is true again when $3 + \alpha$ divisible by 7, i.e.

when α divided by 7, remainder is 4.

30. Let R be a relation from the set {1, 2, 3 ....., 60} to itself such that R = {(a, b) : b = pq, where p, q $\geq$ 3 are prime numbers.}. Then, the number of elements in R is : **[JEE(Main)July2022]**

माना समुच्चय S ={1, 2, 3 ....., 60} हैं। माना S से S में एक संबंध R, R = {(a, b) : b = pq, जहाँ p, q $\geq$ 3 अभाज्य संख्याएँ हैं} है। तो R में अवयवों की संख्या है : **[JEE(Main)July2022]**

(A) 600 (B) 660 (C) 540 (D) 720

Ans. B

Sol. Number of possible values of a = 60, for b = pq,

If p = 3, q = 3, 5, 7, 11, 13, 17, 19

If p = 5 q = 5, 7, 11

If p = 7 q = 7

Total cases = 60 × 11 = 660

31. Let A = {1, 2, 3, 4, 5, 6, 7}. Define B = {T ⊆ A : either 1 ∉ T or 2 ∈ T} and C = {T ⊆ A : T the sum of all the elements of T is a prime number}. Then the number of elements in the set B ∪ C is _____.

[JEE(Main)July2022]

माना A = {1, 2, 3, 4, 5, 6, 7} है। समुच्चय B = {T ⊆ A : या तो 1 ∉ T या 2 ∈ T} तथा C = {T ⊆ A : T के सभी अवयवों का योगफल एक अभाज्य संख्या है} हैं। तो समुच्चय B ∪ C में अवयवों की संख्या है _____

[JEE(Main)July2022]

Ans. 107

Sol. ∵ (B ∪ C)' = B' ∩ C'

B' is a set containing sub sets of A containing element 1 and not containing 2.

And C' is a set containing subsets of A whose sum of elements is not prime.

So, we need to calculate number of subsets of

{3, 4, 5, 6, 7} whose sum of elements plus 1 is composite.

Number of such 5 elements subset = 1

Number of such 4 elements subset = 3 (except selecting 3 or 7)

Number of such 3 elements subset = 6 (except selecting {3, 4, 5}, {3, 6, 7}, {4, 5, 7} or {5, 6, 7})

Number of such 2 elements subset = 7 (except selecting {3, 7}, {4, 6}, {5, 7})

Number of such 1 elements subset = 3 (except selecting {4} or {6})

Number of such 0 elements subset = 1

$n(B' \cap C') = 21 \Rightarrow n(B \cup C) = 2^7 - 21 = 107$

32. Let A = {1, 2, 3, 4, 5, 6, 7} and B = {3, 6, 7, 9}. Then the number of elements in the set {C ⊆ A : C ∩ B ≠ ϕ} is _____. **[JEE(Main)July2022]**

माना A = {1, 2, 3, 4, 5, 6, 7} तथा B = {3, 6, 7, 9} हैं। तो समुच्चय {C ⊆ A : C ∩ B ≠ ϕ}में अवयवों की संख्या है _____. **[JEE(Main)July2022]**

Ans. 112

Sol. A = {1,2,3,4,5,6,7} and

B = {3,6,7,9}

Total subset of A = $2^7 = 128$

C ∩ B = ϕ when set C contains the element 1, 2, 4, 5

∴ S = {C ⊆ A; C ∩ B ≠ ϕ}

= Total . (C ∩ B = ϕ)

= $128 - 2^4 = 112$

33. Let S = {4, 6, 9} and T = {9, 10, 11, ..., 1000}. If A = $\{a_1 + a_2 + ...+a_k : k \in N, a_1, a_2, a_3, ..., a_k \in S\}$, then the sum of all the elements in the set T – A is equal to _____. **[JEE(Main)July2022]**

माना S = {4, 6, 9} तथा T = {9, 10, 11, ..., 1000} हैं। यदि A = $\{a_1 + a_2 + ...+a_k : k \in N, a_1, a_2, a_3, ..., a_k \in S\}$ तो समुच्चय T – A में सभी अवयवों का योग है _____ **[JEE(Main)July2022]**

Ans. 11

Sol. S = {4,6,9} T = {9,10,11..1000}

$A\{a_1 + a_2 +.....+ a_k : K \in N\}$ & $a_i \in S$

Here by the definition of set 'A'

$A = \{a : a = 4x + 6y + 9z\}$

Except the element 11, every element of set T is of of the form $4x + 6y + 9z$ for some $x, y, z \in W$

$\therefore$ T – A = {11}

34. Let $S = \{(x, y) \in N \times N : 9(x-3)^2 + 16(y-4)^2 \le 144\}$ and $T = \{(x, y) \in R \times R : (x-7)^2 + (y-4)^2 \le 36\}$. Then n (S $\cap$ T) is equal to _____. **[JEE(Main)July2022]**

माना $S = \{(x, y) \in N \times N : 9(x-3)^2 + 16(y-4)^2 \le 144\}$ तथा $T = \{(x, y) \in R \times R : (x-7)^2 + (y-4)^2 \le 36\}$ हैं। तो n (S $\cap$ T) बराबर _____ है। **[JEE(Main)July2022]**

Ans. 27

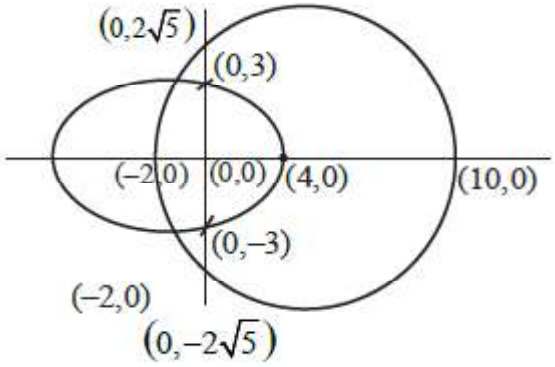
Sol. $S : \frac{(x-3)^2}{16} + \frac{(y-4)^2}{9} \le 1; x, y \in \{1, 2, 3, \ldots\ldots\}$

$T : (x-7)^2 + (y-4)^2 \le 36 x, y \in R$

Let $x - 3 = x : y - 4 = y$

$S : \frac{x^2}{16} + \frac{y^2}{9} \le 1; x \in \{-2, -1, 0, 1, \ldots\ldots\}$

$T : (x-4)^2 + y^2 \le 36 \;;\; y \in \{-3, -2, -1, 0, ...\}$



S $\cap$ T = (–2, 0), (–1, 0), ...(4, 0) $\rightarrow$ (7)
(–1, 1), (0, 1), ...(3, 1) $\rightarrow$ (5)
(–1, –1), (0, –1), ...(3, –1) $\rightarrow$ (5)
(–1, 2), (0, 2), (1, 2), (2, 2) $\rightarrow$ (4)
(–1, –2), (0, –2), (1, –2), (2, –2) $\rightarrow$ (4)
(0, 3) (0, –3) $\rightarrow$ (2) .....